NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics : Capstone 2

Brazilian E-Commerce Analytics

Order Fulfilment, Delivery Performance & Customer Satisfaction (2016–2018)

Sector	E-Commerce / Retail
Team	Section B • Group G-7
Faculty Mentor	Satyaki Das
Dataset	Olist Brazilian E-Commerce (Kaggle, 2016–2018)

Team Members

Saumya Kumar | Saman Iqbal Ansari | Arina Ali | Yash Kumar | Ayush Sahu | Prakhar Srivastava

GitHub: github.com/oksaumya/SectionB_G7_BrazilianE-Commerce

Tableau Public: https://public.tableau.com/app/profile/saman.iqbal5804/viz/BrazillianEDA/SalesOverview?publish=yes&showOnboarding=true

Context & Problem Statement

Why this matters and what decision the analysis supports

SECTOR CONTEXT

Brazilian e-commerce is growing fast on the back of smartphone penetration and expanding logistics but the country's geography and multi-vendor model create wide variance in delivery quality. Olist, Brazil's largest marketplace, sits at the centre of this tension between scale and reliability.

DECISION-MAKER

COO and Head of Seller Management of a large Brazilian online marketplace. They need to prioritise logistics investment, set seller onboarding standards, and decide where to spend customer-retention budget first.

CORE BUSINESS QUESTION

Which products, sellers, regions, and delivery patterns should management focus on to reduce service issues and improve revenue and repeat business?

OBJECTIVE

- Identify the operational drivers of late delivery, complaints, and revenue leakage.
- Quantify their impact and translate findings into 3–5 actionable recommendations the COO can fund.

Data Engineering — Source to Pipeline

From nine raw Olist tables to a single 100K-row analysis-ready master

SOURCE

- **Dataset:** Olist Brazilian E-Commerce (Kaggle)
- **Volume:** ~114,092 orders, 9 CSVs
- **Shape:** 100K+ rows × 37 cols (post-merge)
- **Period:** Sep 2016 to Aug 2018
- **Schema:** Relational, joined on order_id / customer_id / seller_id / product_id

PYTHON ETL (notebooks 01 & 02)

- **Nulls:** delivered_date 3.1% retained as NaN; review_score 0.8% excluded; category 0.1% → 'Unknown'
- **Outliers:** Negative delivery durations dropped; >$5K orders kept (legit bulk)
- **Types:** All timestamps → datetime64; UF state codes standardised
- **Engineering:** delivery_days, delivery_delay_days, late_flag, order_month/quarter/year
- **Merge:** pd.merge() across 8 tables → olist_cleaned_master.csv

GITHUB & DICTIONARY

```
notebooks/
  01_extraction.ipynb
  02_cleaning.ipynb
  03_eda.ipynb
  04_statistical_analysis.ipynb
  05_final_load_prep.ipynb

data/
  raw/   (9 CSVs)
  processed/
    cleaned_master.csv
```

KEY COLUMNS (analysis surface)

Column	Role	Column	Role
order_id	join key	review_score	satisfaction KPI
order_purchase_timestamp	time trends	payment_value	revenue
order_delivered_customer_date	delay calc	payment_type	payment behaviour
order_estimated_delivery_date	delivery gap	product_category_name	category perf.
order_status	fulfilment	seller_state / customer_state	geographic split

KPI & Metrics Framework

Five KPIs that map directly to the COO's decision levers

$18.1M	103.43K	$175	4.03 / 5	92.3%
Total Revenue 2016–2018	Total Orders delivered	Avg Order Value	Avg Review Score	On-Time Delivery Rate

DEFINITIONS & FORMULAS

KPI	Why it matters	Formula	Notebook
Monthly Revenue Growth %	Tracks platform momentum	(Cur − Prev) / Prev × 100	05_final_load_prep
Avg Customer Review Score	Customer satisfaction proxy	mean(review_score)	03_eda
Late Delivery Rate	Fulfilment failure exposure	late_flag=1 / delivered × 100	04_statistical
Cancellation Rate	Revenue leakage	cancelled / total × 100	03_eda
On-Time Delivery Rate	Service-quality baseline	on_time / delivered × 100	04_statistical

WHY THESE FIVE

Each KPI maps to one lever the COO actually controls growth, satisfaction, logistics, and revenue leakage — so every chart on the dashboard answers a question someone in the room can act on.

Key Insights from EDA

Six findings stated in business language — supporting charts in 03_eda.ipynb

1. Revenue grew ~4× year-on-year, but is concentrated.

Q1 2018 hit $3.74M vs Q1 2017's $0.94M. Top-3 categories (bed/bath/table, computers, health/beauty) drive a disproportionate share disruption in any one materially impacts platform revenue.

2. São Paulo dominates both sides of the marketplace.

SP customers spend $7.55M; SP sellers earn $11.87M. Any uniform policy hits SP first and hardest SP must be modelled as a separate operating regime.

3. Late deliveries hit satisfaction non-linearly past day 13.

Fast (<8d) reviews cluster above 4.0; slow (>13d) reviews fall below 3.0 with a steeper slope each extra delay day costs more at higher delays.

4. Customers pay in instalments: credit card = 76.8% of revenue, ~3 instalments avg.

Revenue at order time ≠ cash realised; cash-flow planning for sellers depends on instalment lag, not order count.

5. Review distribution is bimodal, not normal.

5★ = 56.4%, but 1★ = 12.8% (13,194 orders) vs only 3,484 at 2★. Dissatisfied customers report extreme dissatisfaction, consistent with delivery failures.

6. Q2 is the highest-revenue quarter every year.

$3.84M in 2017 and comparable in 2018 operations and seller readiness must be scaled into Q1 not after.

Advanced Analysis — Statistical Depth

Going beyond EDA: where is the delay-satisfaction relationship strongest, and who owns it?

DELIVERY TIME vs REVIEW SCORE — segmented regression

- **Fast (<8d):** review scores cluster above 4.0
- **Medium (8–13d):** scores 3.8–4.1, mixed
- **Slow (>13d):** scores below 3.0, steepest slope
- **Finding:** non-linear each additional day costs more after the 13-day threshold

LATE-ORDER CONCENTRATION — by seller state

- **SP:** 6,730 late orders — 12× the next state (RJ 382)
- **MG / PR:** 481 / 550 late — secondary watch list
- **ES / BA:** 24 / 34 late — low absolute risk
- **Finding:** SP-specific intervention beats nationwide programmes on cost-of-impact

SEASONAL & CATEGORY EFFECTS

Late Order Volume by Month

Jan / Feb / Mar / Nov spikes (1,260 / 554 / 1,812 / 267); mid-year (Jun–Aug) shows lowest late counts. Avg delivery days drift 9.4 → 14.8 across the year — holiday-cycle pressure is visible.

Delivery Performance by Category

Audio, agro-industry, and art categories sit materially above the 12-day platform average — heavier or specialist logistics. Lightweight craft items deliver fastest.

Methods used: pandas + numpy aggregation, scipy/statsmodels for trend lines and segment comparisons, matplotlib/seaborn for visual validation — all in 04_statistical_analysis.ipynb.

Tableau Dashboard — Walkthrough

Three pages, six global filters, KPI-driven layout for executive and operational audiences

Page 1 — Sales Overview

Executive view: $18.1M revenue, 103K orders, monthly trend, top categories.

Page 2 — Customers & Geography

86.88K unique customers, SP dominance, state-level revenue, review distribution.

Page 3 — Delivery & Operations

92.3% on-time, 8,756 late orders, delay-by-category, review vs delivery scatter.

GLOBAL FILTERS (live across all 3 pages): Year | State | Order Status | Payment Type | Product Category | Quarter

TABLEAU PUBLIC URL

[https://public.tableau.com/app/profile/saman.iqbal5804/viz/BrazillianEDA/SalesOverview?publish=yes&showOnboarding=true]

Business Recommendations

Five prioritised actions each linked to an insight and an estimated impact

1

SP Seller Compliance Programme

Insight: Insight 5: SP sellers = 76.8% of late orders.

Action: Tiered scorecard; <90% on-time triggers review; <85% × 2 months → listing restrictions.

EXPECTED IMPACT

−673 late orders/period; +0.1–0.2 review-score points

2

Better Delivery Estimates + Proactive Comms

Insight: Insight 3: ~9-day average overrun on late orders.

Action: Re-fit estimate algorithm by state×category; auto-notify customers at 80% of window.

EXPECTED IMPACT

15–20% reduction in expectation-driven 1-star reviews

3

Top-3 Category Logistics SLA

Insight: Insight 1: top-3 categories drive disproportionate revenue & delays.

Action: Dedicated carrier SLAs + regional inventory pre-positioning for those categories.

EXPECTED IMPACT

$200K–$400K at-risk revenue protected per year

4

1-Star Customer Recovery Campaign

Insight: Insight 5b: 13,194 1-star reviewers — high churn risk.

Action: 48-hour personalised recovery: apology + voucher + expedited resolution; track 90-day repurchase.

EXPECTED IMPACT

~$346K incremental revenue (15% recovery × $175 AOV)

5

Pre-Q2 Demand Scaling

Insight: Insight 6: Q2 is peak, but Jan–Mar already spike late orders.

Action: Inventory readiness check 6 weeks pre-Q2; pre-negotiated carrier capacity for Jan–Mar.

EXPECTED IMPACT

≈350 fewer late orders per Q1–Q2; on-time rate → 93.7%

Impact & Value — The 'So What?'

What the platform stands to gain and what it loses by waiting

AGGREGATE IMPACT (low–high range, conservative assumptions)

$546K+	≈1,000	+0.1–0.2	–15–20%
Combined revenue at risk protected or recovered annually	Fewer service failures per Q1–Q2 cycle	Points lift in average review score	Reduction in expectation-driven 1-star reviews

WHY ACT NOW

- **Compounding churn:** 1-star reviewers who don't return cost ~$175 AOV × repeat-order multiplier each and Q1 alone has ~3,000 of them.
- **Concentration risk:** 76.8% of late orders sit with SP sellers; without intervention, every quarter's growth re-amplifies the same bottleneck.
- **Cheap wins are real:** Recommendation 2 (estimate accuracy) requires zero logistics investment only an algorithm and notification fix yet captures 15–20% of expectation-driven complaints.
- **Cost of inaction:** Even a 0.3-point review-score drop typically halves repeat-purchase probability; on $18.1M base, the CLV exposure is in the multi-$M range over 2 years.

Numbers are approximate; logic is documented in §11–12 of the project report.

Limitations & Next Steps

What we couldn't conclude and what should come next

LIMITATIONS

- **Correlation, not causation:** delivery vs review is observational; product quality, packaging, and seller comms are not controlled for.
- **Repeat-purchase weakness:** customer_id is per-order, not persistent robust repeat-rate computation requires deduplication via customer_unique_id.
- **Seller anonymisation:** no qualitative seller data (size, age, reputation) we cannot diagnose why a seller underperforms.
- **Time truncation:** data ends Aug 2018 - no full-year 2018 view, no late-2018 holiday peak.
- **Order-level granularity:** no product-level margin or return rate unit economics out of scope.
- **Single platform:** findings apply to one marketplace; generalisation to Brazilian e-commerce broadly is not warranted.

NEXT STEPS

- **Predictive late-delivery model:** logistic regression / GBM on state × category × payment × timing for risk scoring at order placement.
- **Seller segmentation:** K-means on (on-time rate, review, revenue, volume) → tiered support programmes.
- **Review-text NLP:** sentiment + topic analysis on review comments to surface specific complaint drivers.
- **Real-time dashboard:** API-connected pipeline → operational alerting on emerging delay spikes.
- **Customer Lifetime Value:** link first-delivery experience to long-term repeat behaviour; quantify CLV loss per late delivery.

Team & Contribution

Visible commits, owned phases, declared in the report's contribution matrix

Member	Data	ETL	EDA	Stats	Tableau	Report	PPT
Saumya Kumar	**Owner**	Support	Support	—	Support	Support	**Owner**
Saman Iqbal Ansari	Support	**Owner**	**Owner**	Support	**Owner**	—	—
Arina Ali	Support	**Owner**	—	—	Support	**Owner**	Support
Yash Kumar	—	—	—	Support	**Owner**	Support	—
Ayush Sahu	—	—	—	—	—	—	Support
Prakhar Srivastava	—	Support	**Owner**	Support	—	—	—

PROJECT METADATA

```
Team Lead:    Saumya Kumar
Mentor:       Satyaki Das
Institute:    Newton School of Technology
Section:      B
Group:        G-7
Date:         April 2026
```

EVIDENCE TRAIL

- GitHub Insights: commit history per member
- Pull-request log: ownership of each phase
- Contribution matrix declared in §B of report
- All notebooks committed in .ipynb format

Thank You

Questions?

ONE-LINE TAKEAWAY

The Olist platform's growth story is real but operational reliability, concentrated in São Paulo sellers and a handful of categories, is the single highest-leverage decision surface for the COO over the next year.

GitHub: github.com/oksaumya/SectionB_G7_BrazilianE-Commerce

Tableau Public: https://public.tableau.com/app/profile/saman.iqbal5804/viz/BrazillianEDA/SalesOverview?publish=yes&showOnboarding=true